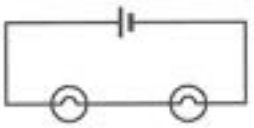
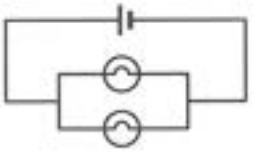
Series and parallel circuits 5a

A series circuit means that all components are connected end to end in the same circle, that is there is only one pathway. In a parallel circuit, electrons are given a choice of an alternative route to get around the circuit; in other words they form two or more circles off the same power source. The other routes are called branches.

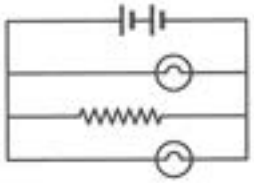
1 Label the diagrams below as either *in series* or *parallel* circuits.



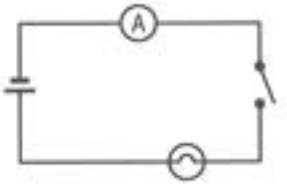
a _____



b _____



c _____



d _____

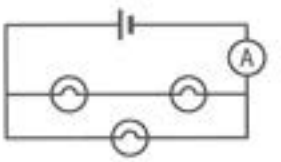
2 Match the descriptions to the diagrams by drawing lines between texts and diagrams.

a This circuit is made up of a 12 volt battery that is powering three light globes in series. An ammeter is connected to measure the current going through the light globes.

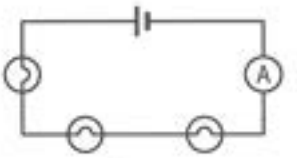
b This circuit is made up of a 12 volt battery that is powering three light globes in parallel. An ammeter has been connected to measure the total current flowing through the light globes.

c This circuit is made up of a 12 volt battery powering three light globes, two of which are in series, with the third one in parallel. An ammeter is connected to measure the current entering the light globes.

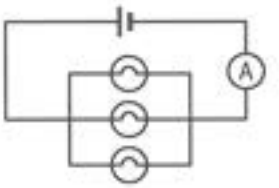
A



B



C



3 Draw the following circuit diagrams.

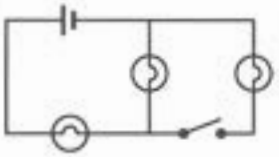
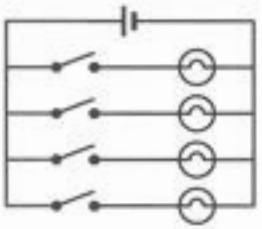
a A 6 volt battery powering two lights and a variable resistor in parallel, with each light having its own switch.

b A 12 volt battery powering two globes in series. The circuit also has an ammeter and a switch at one end of each globe.

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Series and parallel circuits 5b

4 Write a description for each of these circuit diagrams.

Circuits	Descriptions
a 	
b 	

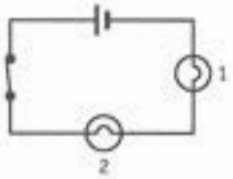
5 If one globe in a series circuit was to go out, the other globes would go out too. If one globe in a parallel circuit was to go out, the other globes in the circuit would stay lit. Why?

_____

_____

_____

6 Observe the diagram to the right then answer the following questions.



a What kind of circuit is this? _____

b How many possible paths can the electrons follow? _____

c Electrons move from the negative terminal to the positive terminal. Draw arrows on the diagram to show the path of the current.

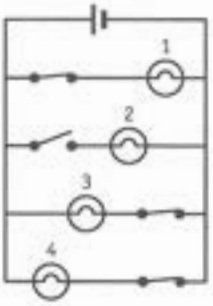
d If globe 1 was removed, what would happen to globe 2? _____

_____

e If globe 1 was replaced with a wire, what would happen to the brightness of globe 2?

_____

7 Observe the diagram to the right, then answer the questions.



a What kind of circuit is this? _____

b How many paths can the electrons follow? _____

c If globe 2 was removed, what would happen to globes 1 and 3?

_____

d Looking at the diagram, which globe is not working?

Why? _____

e Describe the brightness of each globe. _____

_____

8 Do you think your house would be wired like the diagram in question 6 or 7? Give a reason for your answer. _____

_____

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